Better Gardens Prototype

Fundamentals of Interaction Design INFO10003

Our online **Figma prototype** is at: [Figma Prototype]

Our **presentation video** is at: [YouTube URL]

Abstract

After researching gaps in the current community garden resources it was found that there is currently no app that includes all necessary features specifically for a multigenerational demographic. The aim of this study was to create a simple multifunctional application which did not overwhelm a mature experienced user and supported a younger user with little gardening experience. We conducted the same data collection process with 5 different participants all with different gardening experience, technical experience, and involvement in community gardens found from the University of Melbourne. Using data from assessment 1 to make a low fidelity application design and creating a high fidelity functional design on figma for all participants to test and give feedback by recording them going through the app and timing how long it took to complete each task given to them. We asked them to complete the standard version of the system usability scale questionnaire. This feedback was applied to improving the high fidelity design. The final design of better gardening will aid in better functioning community gardens, helping beginner gardeners, and diverse participants in community gardens.

1. Introduction

Our prototype of a gardening app, ‘Better Gardens’ is a building on the past ideas developed through interviews and field research. The overall goal was to connect people through a multigenerational system that eradicates the assumption that only older people enjoy the act of gardening. After the development of a low-fidelity functional prototype, more research was conducted to gather data that ultimately led to the final high-fidelity design. This gathering of data showed us holes in our design as well as gave feedback for improvements that we were able to take in and apply to our prototype.

2. Background work

Throughout assignment one, an exploration into the current available resources and how they are able to inform our design was conducted. The collection of the research reports displayed a key theme of the

social and environmental implications of utilising the tool of technology within a community garden setting. Conclusions such as an overall increase in a person's physical and mental health and community connections were drawn, as well as reports of the availability of technology only being a useful resource to a point. Along with academic research being conducted, peer feedback was collected. From these interviews, we were able to gather valuable insights into what a real peer would value in a resource within the correct context. Some points found were that, throughout the younger audiences, a higher focus on the help of technology was established, contrasting with an older audience who presented contrary views. Along with this, both audiences were unanimous on the idea that reminders would be a useful tool.

After these conclusions were drawn, they went on to inform the design of the initial prototype. A heavy focus on the inclusion of reminders was at the forefront, presenting as the predominant feature, following on from this, plant information and social interactive elements, such as a map, a notice board and a profile establishment for connection. Gaps in the market consist of an application that combines all of the features that individual applications possess, along with this, most applications are not location specific, providing a clear opportunity to develop a resource that accounts for the user's context, benefiting both the social and environmental implications of the app

Throughout the third assignment, we aim to further expand on the conducted research in an attempt to produce a community garden tool that benefits the largest audience.

3. Methods

The first thing we did when designing our prototype was look back at the feedback from Assignment One to see what we could improve. We believed our original app design was already pretty solid for a low-fidelity paper prototype, with most features working well and feeling functional. However, there were still a few things we wanted to add or tweak, so we updated that design and used it as the base for our new low-fidelity version for this assignment. From there, we moved into Figma and started building all the screens and pages. Once we had everything laid out, we explored different colour schemes and decided on earthy tones to fit the theme of our app. We kept adjusting contrast and layout along the way to ensure everything felt balanced and clear.

There are nine pages, including the login, home, plants, tasks and reminders, calendar, community page, which includes messages, a map page, and a profile page. This high-fidelity design was made

functional on Figma for our participants to test. The same participants from assignment one were used in the testing process. Each interview was recorded, and each participant was given a list of tasks to complete on the application. They were asked to log in, add a new community garden to the home page, get more information on the basil plant, and switch the task page from to-do list mode to calendar mode. Participants were asked to talk about what they were doing and thinking during these tasks. During this interview, we timed how long each task took to complete. We used a standardised questionnaire called the standard system usability scale. This questionnaire consists of 10 questions where the participants can rank on a scale of strongly disagree to agree with each statement strongly. We then asked for general feedback on what the participants thought could be improved and what worked for them. Once all interviews were conducted, we read through all feedback, timing for each task, and the average ranking from the questionnaire. We collectively decided on what to edit in our design.

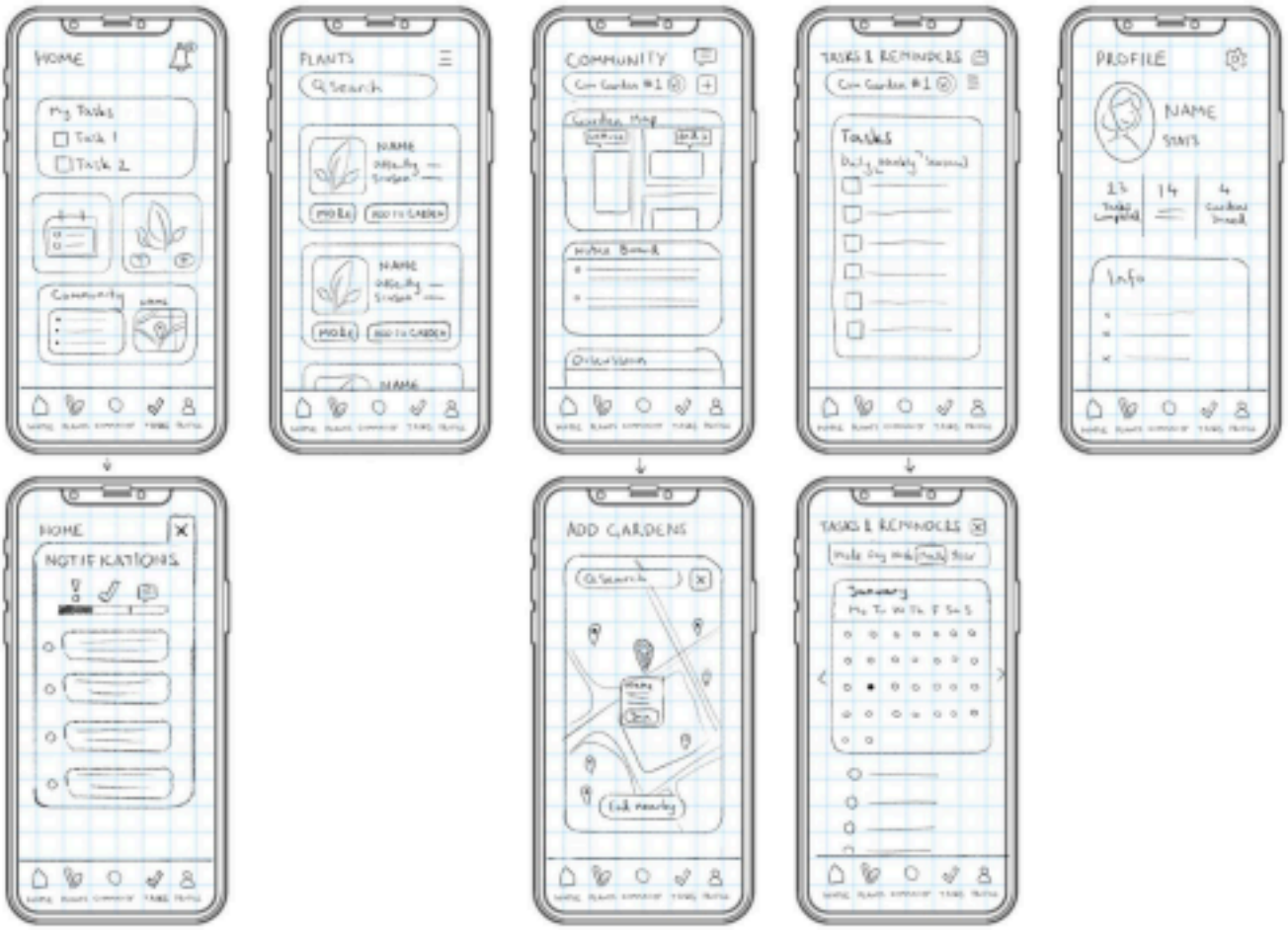
Table 1. Table of participants

	Gender	Age	Occupation	Experience with gardening	Technological ability
Participant 1	Female	21	Student	No Garden Experience	Uses computer and phone on a daily basis, and has advanced technology abilities
Participant 2	Female	57	Architect	Avid home gardener	Uses computer applications for work and recreation, but not completely proficient in all technology types.

Participant 3	Male	19	Student	Slight gardening experience.	Uses computers and technology almost everyday. Very proficient at all technology, and also works in a technology company.
Participant 4	Female	57	Lecturer	Avid home gardener	Uses computers and applications like Canvas daily, but not as proficient on applications that she is unfamiliar with.
Participant 5	Female	19	Student	Basic gardening skills	Has proficient knowledge of many technological applications

4. Prototype

4.1 Paper Prototype (low-fidelity)



Home Page:
The home page includes widgets that act as shortcuts to other parts of the app.

- A **task list widget** lets users quickly check off tasks and links to the full task list.
- A **calendar widget** leads to the detailed calendar view on the Tasks page.
- A **plant widget** automatically scrolls through featured plants, with shortcuts to “More Info” and “Add to Garden” on the Plants page.
- A **community widget** with shortcuts leading to the different community page boards, and the community garden map.
- A **notifications button** opens a pop-up modal that can be filtered by reminders, tasks, or

messages.

Plants Page:

This page uses a scrollable feed (similar to TikTok) displaying different plants. Each card includes options to get more information or add the plant to a garden. Users can also filter plants by difficulty and season to suit their gardening level and timing.

Community Page:

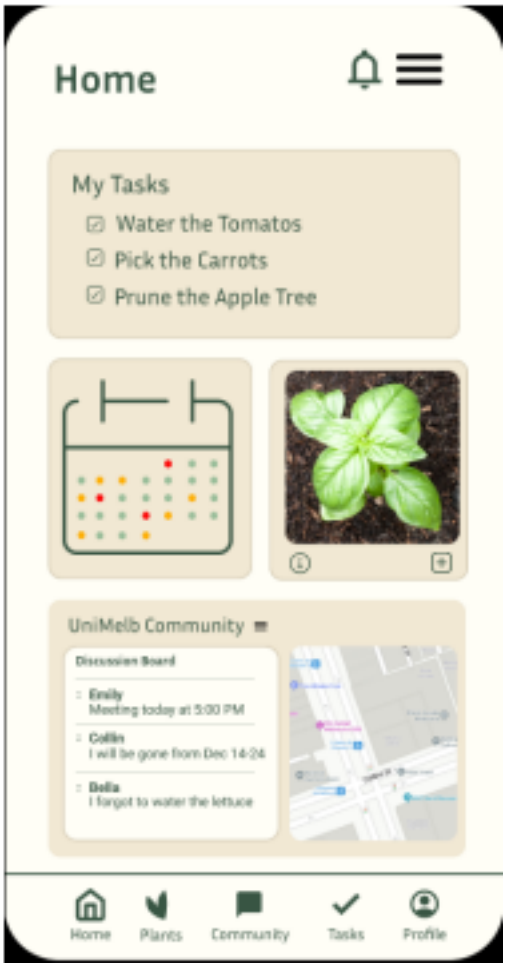
Users can switch between different community gardens they are part of. An Add Gardens button opens a map where users can find and join nearby gardens. The page also includes a garden map, notice board, and discussion area allowing communication between the community garden members.

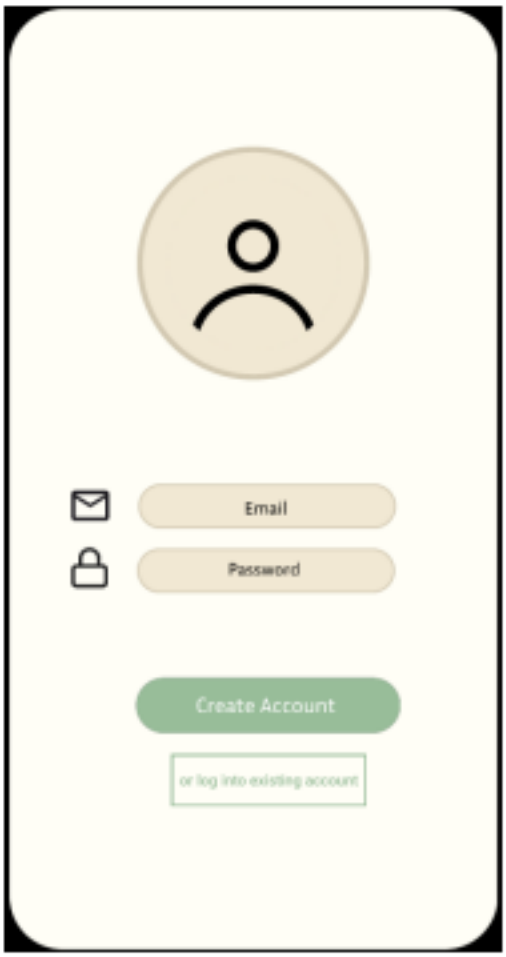
Tasks and Reminders Page:

Users can toggle between List Mode and Calendar Mode.

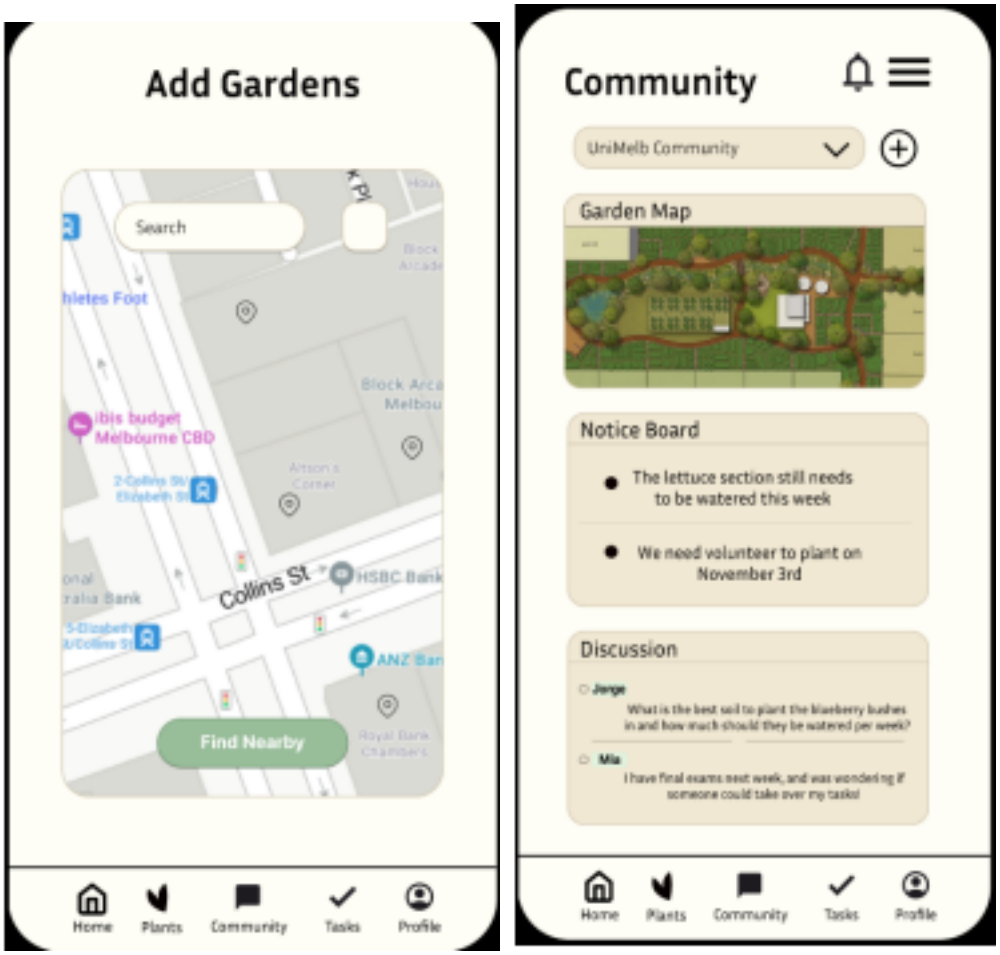
- In **List Mode**, tasks can be filtered by day, week, or season, and checked off once completed. ● In **Calendar Mode**, tasks are displayed in a colour-coded calendar which indicate the task's urgency and scheduling.

4.2 Digital Prototype (pre-evaluation)

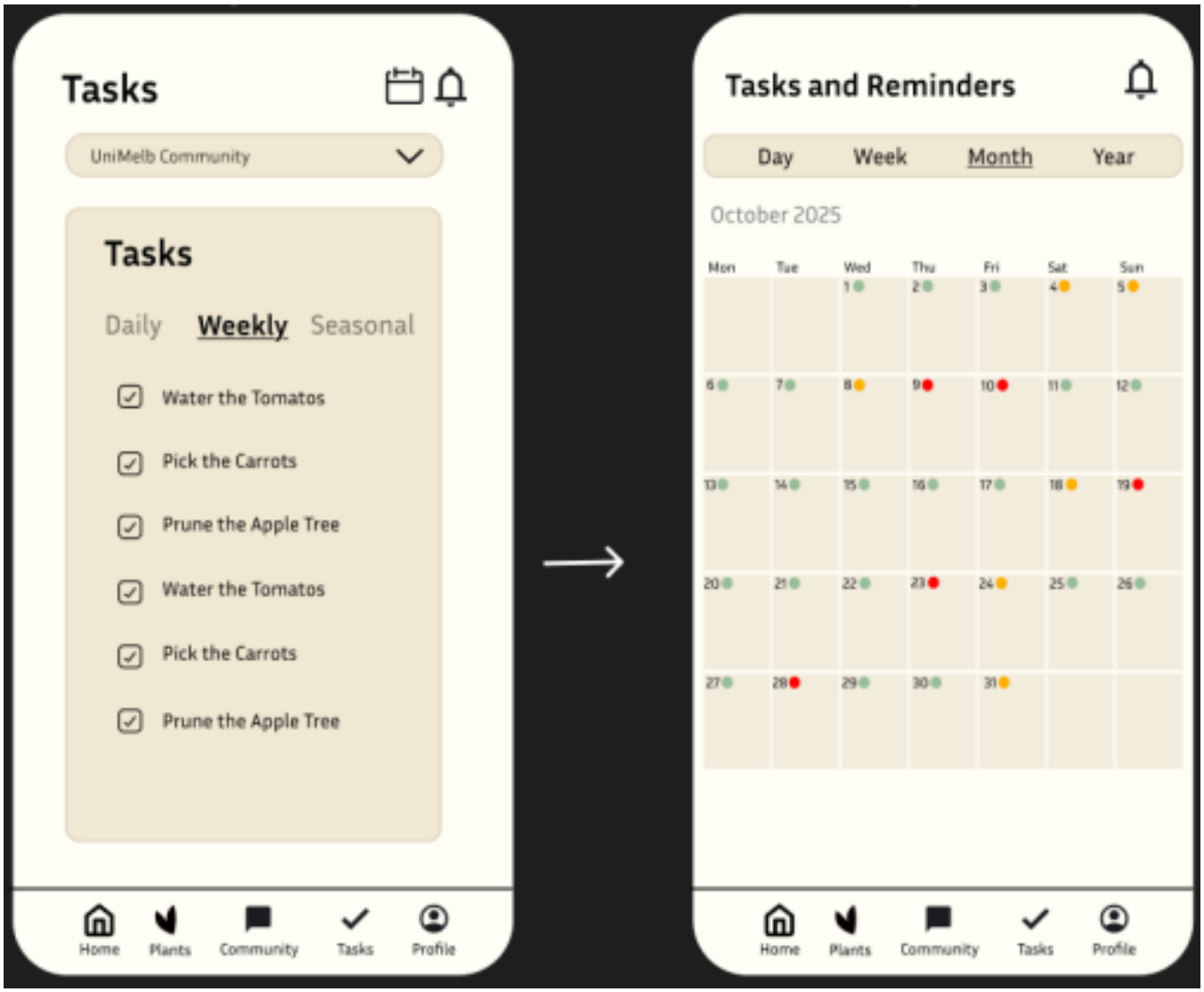




The prototype begins on the login page. By clicking “Create Account” or “Log Into Existing Account”, the user is directed to the home page. On the home page, six interactive widgets provide shortcuts when clicked. Additionally, the navigation bar at the bottom of the screen is fully interactive and allows the user to navigate to different pages.



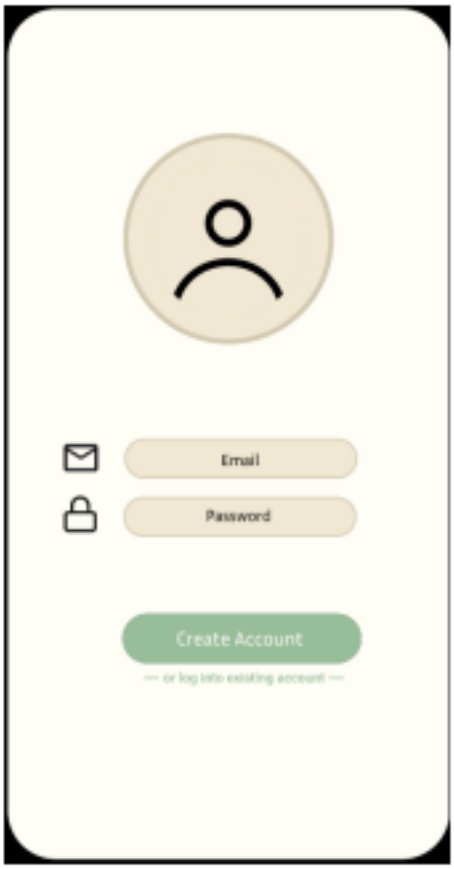
On the community page, users can see the current garden community boards, and by pressing the plus icon, they can access the feature of adding a new garden nearby.



Users can view their tasks in a to-do list format on the task page. Alternatively, by clicking on the calendar icon, they can display their tasks in a calendar view, with different levels of urgency indicated by colors (green, orange, and red).

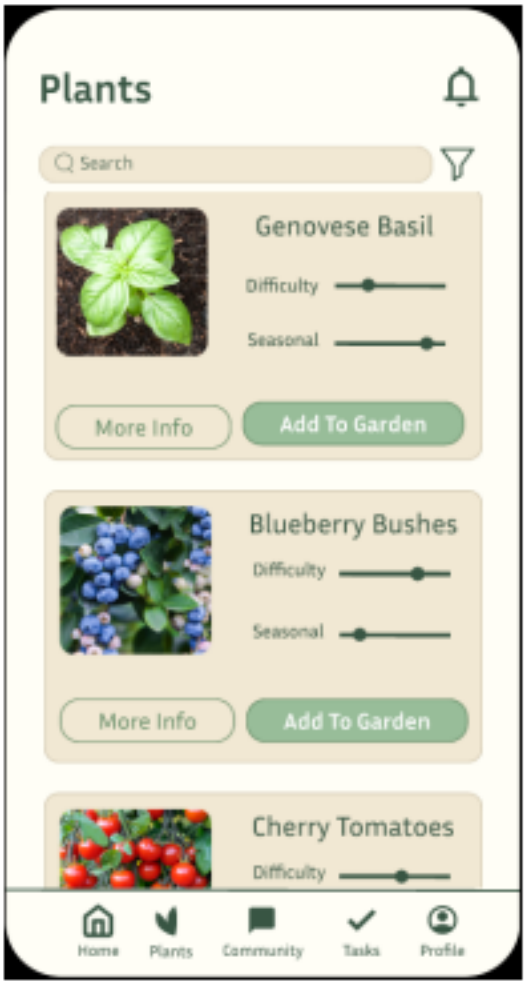
4.3 Digital Prototype (post-evaluation)

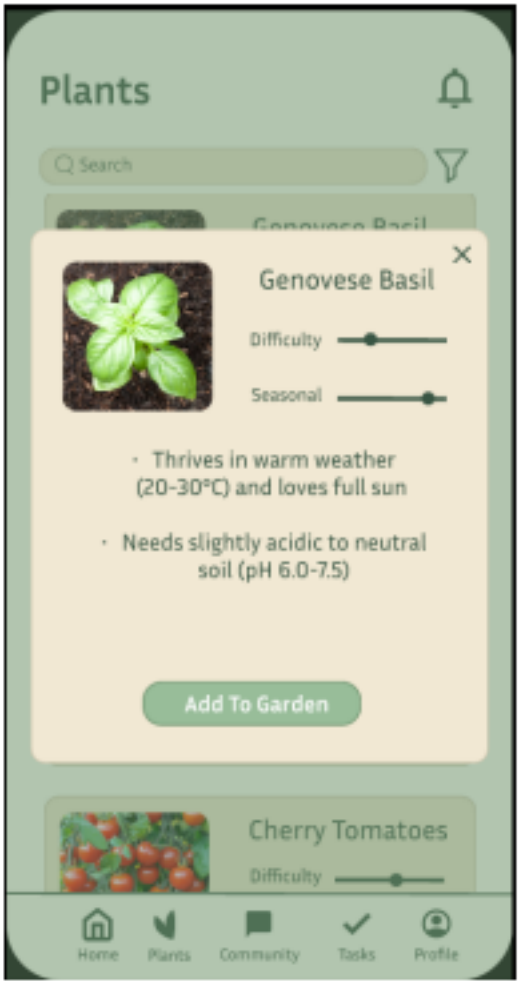

Profile
Sabrina Floyd
@swb_floyd
Joined the Community Gardens in July 2025
Stats
23
Tasks Completed
7
Discussion Contributions
4
Gardens Joined
Interests
Landscape designs and garden layouts
Growing herbs and vegetables
Garden journaling and plant tracking
Home
Plants
Community
Tasks
Profile


Email
Password
Create Account
— or log into existing account! —

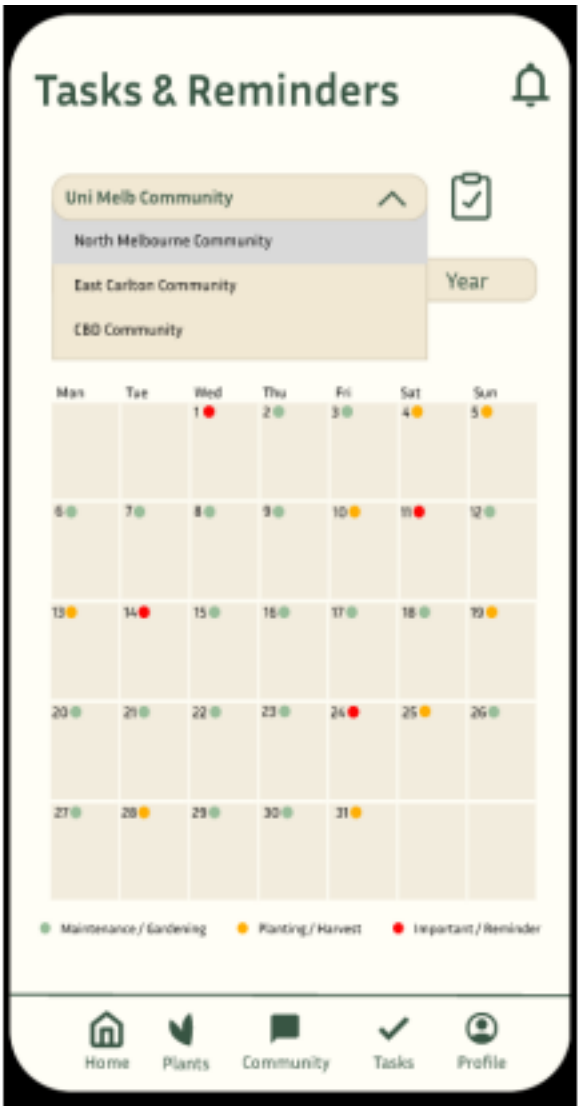


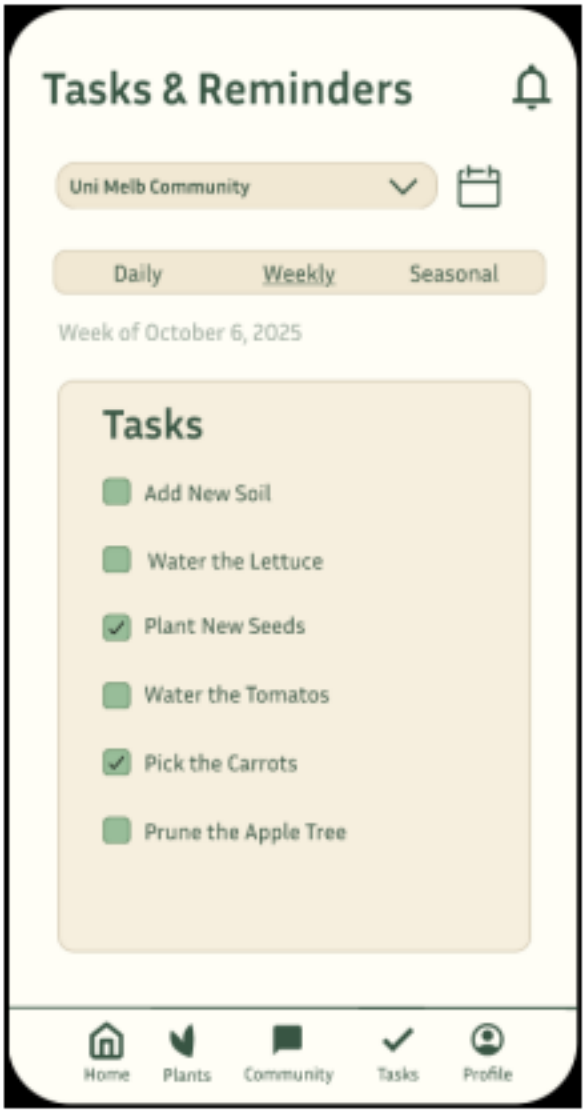
The design and features of the login and home pages remained consistent before and after the evaluation. Users can still access the home page by logging into an existing account or creating a new one. On the home page, the “My Tasks” widget is now interactive, allowing users to check off completed tasks or let them remain unchecked. The six widgets continue to serve as shortcuts to different sections of the app and the navigation bar makes it easy for users to access different pages.





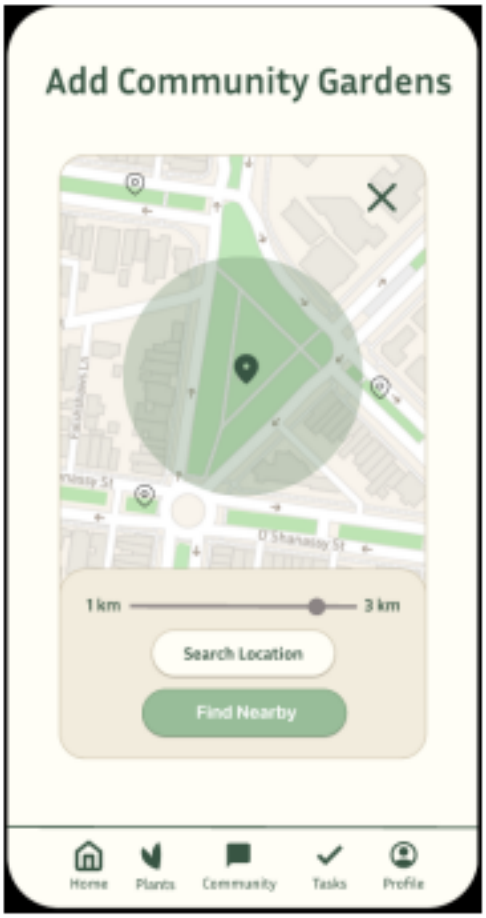
The plants page underwent several changes following the evaluations. Prior to the evaluation, it did not include any interactive features. However, the team identified key additions we planned to include in the low-fidelity stage that could enhance the user experience. Users can now scroll through various plants in a TikTok-style feed on the updated Plants page and click the “more info” button to expand detailed information about a specific plant and decide whether to add it to their community garden.





The tasks page includes several key features with additional new features to enhance interactivity and complete previously underdeveloped features. The same functionality on the home page under “My Tasks” is also included here. By clicking on boxes following each task, the user can tick/check off different tasks they have completed. Additionally, by clicking on the calendar icon, the user can switch to a calendar mode, and now they can return to the to-do list view by clicking the checklist board icon. Another new feature is the dropdown menu, allowing users to switch between community gardens to see the different tasks and reminders.





On the community page, users can view relevant notices, discussion boards, and a map of the community garden. This page also allows users to add their first or an additional community garden by clicking “Add Community Garden”. Similar to the tasks and reminders page, the user can now switch between the different community gardens to view information specific for each community.

5. Results

5.1 Qualitative

Likes	Dislikes
- Straightforward and easy to use	- Did not feel like a fluid system and had incomplete features
- Felt confident about switching between the different pages	- Many features/ buttons lead to the same pages and resources

- Simple design and wasn't too overwhelming	- Hard to find exactly what they were looking for (eg. missing filters tab and specific searches)
- Concept of the app feels unique	- "Add gardens" feature was unclear and couldn't understand how it was supposed to be used
- NNavigation bar at the bottom with good imagery	- Didn't have many features added yet and they appeared to be undeveloped
- Could quickly go through the tasks	- Would like to be able to edit their profile a bit more
	- Lacked consistency throughout many pages. Such as the spacing, padding and placement.
	- The profile page looks very incomplete, and lacks a nice design.

6. Discussion

Table 2. Design changes made following evaluations of our initial Figma prototype

Initial design feature (pre-evaluation)	**Updated design feature (post-evaluation)**	**Justification for change (refer to specific evaluation results here)**
A plus sign was used to indicate the 'add garden feature'	A button that says 'add community garden '	Due to confusion when navigating the prototype, users were confused as to what the specific function of the plus sign was. The update to a button that states 'add community garden' signposts what the feature is able to offer to the user.

Switching from check list view to calendar view with the calendar icon on tasks page	Switching between check list view to calendar view on Tasks and Reminders page by clicking on the corresponding icons (calendar and check list)	Before the post-evaluation, users could switch to calendar mode from the to-do list, but there was no convenient way to switch it back to the to-do list from calendar mode. Adding both icons now allows users to easily switch between the two views.
Find Garden feature shows user gardens available in the nearby areaMultiple features lead to the same page	Added a distance radar (in relation to the users location) and allow users to find gardens in areas they searched forGot rid of certain prototype flows and buttons leading to the same page	Users were confused on how the gardens were found as well as the location of the garden. The radar reduces the confusion as well as allow users to see and find gardens within their desired distance. Users are also able to find gardens from other locations.It made the interface overcomplicated, and having every button and board bringing users to the same place felt redundant.

A bar with an arrow down (representing the potential to have multiple community gardens to switch between)	Interactive drop down menus for Community and Tasks and Reminders	Evaluation showed that this was an underdeveloped and an unfinished feature. It highlighted how this could be a key feature and would incorporate the idea of having multiple community gardens.

Looking for more info about specific plants	Making the “more info" button on the plant page interactive and expandable	During the user study, the users appeared confused when they clicked the button and were unable to access more information about the plants.

After reviewing our evaluation results, we made several updates to improve how our app works and feels. The key changes are mentioned in the table above, but overall, we focused on making things clearer and interactive. We also focused a lot on the visual design, as a lot of the qualitative feedback was on the app’s overall look and aesthetic. We included smaller changes like animations, consistency, spacing, and padding throughout the app. These changes made the app feel much nicer and easier to use, compared to the pre-evaluation version, which lacked a lot of functionality, interactivity, and aesthetics.

A significant limitation of our study was definitely the small number of participants. Since three were university students, and the other two were much older, it did not fully represent the multigenerational audience for which we designed the app. Such little interviews also meant that our quantitative data was not as helpful, as we had very few people taking the interviews. Another significant limitation of our study was that our prototypes were made in Figma; many of us are very inexperienced with Figma, so we could not include many of the features we had discussed earlier. Figma also made it harder for us to try to include any real functions like live map features, notifications, or other interactive elements.

Looking back, we were overly ambitious with what we wanted to include in our prototype. At the start, we had many ideas for features to make the app more complete, like more interactive community sections, better profile customisation, and smarter reminders and notifications. However, since most of us were still new to Figma, we struggled to bring all of those ideas to life. Because of that, many of the features ended up being much more basic than planned or not fully functional. We could continue working on our design to make it more interactive and flow well. However, this would require more time and experience, which would allow us to refine many of our features further.

Before publishing, we also need to be confident that it works well with people who do not know our system and is easy to understand for all our users. That could mean we would need to do another user

study with a bigger test, have them go through more tasks with our newer prototype, and go through a similar process we just did.

References

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